Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Victoris**
b. Team leader name: Yash Shah (team: Pratyaksh, Krishna)
c. Problem Statement: Sustainability & Social Impact: Second Opinion, a live scam guardian that warns seniors (and every age group) before money moves.

Brief about the idea

Second Opinion is a live scam guardian for the people we love. It listens alongside someone on a suspicious call (speakerphone or a shared Google Meet tab), recognises the coercion pattern behind scams such as the fake-police 'digital arrest', and warns them calmly in English or Hindi **before any money moves**.

It then alerts a family guardian on their own phone, requests a cool-off hold the moment money or an OTP is demanded, and writes a ready-to-file complaint for India's 1930 / cybercrime.gov.in. The same engine checks screenshots, voice notes, pasted messages or a plain question ('is this a scam?'), and extends from seniors to teens and young adults.

Why it matters: scams persuade, they don't hack

- Most losses are payments the victim authorises under pressure on a call, which transaction monitoring cannot see.
- Japan 2025: JPY 142.3B lost to special fraud (+98%); people 65+ bore 59.2% of losses.
- India 2024: INR 1,935 crore lost to 'digital arrest' scams across 1.23 lakh complaints.
- Australia 2025: people 65+ are 17.1% of the population but 26.5% of reported losses.

Theme: Sustainability & Social Impact | **Category:** Social Good

Opportunities

How different is it from any of the other existing ideas?

- Existing tools check transactions after the fact, filter spam numbers, or analyse one uploaded message. Second Opinion works **during the call** and reasons about the coercion pattern (authority + urgency + secrecy + payment request), not keywords, in English, Hindi and Hinglish.
- It **acts**: instant guardian alert, guardian-approved cool-off hold, full-call complaint draft, and it shows the exact quote behind every alert.

How will it be able to solve the problem?

- It interrupts the scam at the one moment it can still be stopped: on the call, before the victim authorises the transfer.
- A family member is pulled in within seconds, breaking the isolation scammers rely on; reporting is instant, feeding India's 1930 rapid-reporting system (which had helped save ~INR 8,189 crore by Dec 2025).

USP of the proposed solution

- Live, multilingual and explainable (quotes + official advisories); human-in-the-loop (the guardian approves); privacy-first (no audio stored).
- Safety actions never wait for an LLM, and **one engine serves every age group**: seniors today, teens (sextortion) and young adults (job / task scams) next.

List of features offered by the solution

1. **Call Guard:** listens to speakerphone, a Google Meet tab or a demo call; live transcript; risk meter Calm / Caution / Warning / STOP.
2. **Signal timeline:** every scam signal with the exact quote and time (authority claim, urgency, secrecy, payment / OTP request, threat ...).
3. **Whisper warning** in English or Hindi and a one-tap 'Hang up & call 1930' button; large, calm, senior-friendly UI.
4. **Deterministic safety actions:** instant guardian alert at Warning; cool-off hold as soon as money or an OTP is asked.
5. **Case agent** (Gemini function calling): official advisory, plain-language guardian summary, early complaint draft.
6. **Family guardian console** on a second phone (QR / pairing code): realtime alerts, approve or decline the hold.
7. **Final complaint draft** for 1930 / cybercrime.gov.in written from the full call: category, timeline, quotes, amount; copy / print.
8. **Scam Check:** chat screenshots, voice notes / call recordings, pasted text or a question; verdict, risk score, red flags, safe reply, follow-up chat.
9. **Frame Check:** video-call frames checked for staged-authority cues (uniform, fake warrant); cues only, never a deepfake verdict.
10. **All-ages profiles:** Senior (core), Adult, Young adult, Teen (beta) change tone and scam taxonomy; Hindi / English.
11. **Impact page:** sourced statistics and an avoidable-loss calculator; **Recorded replay** mode and a 16-call evaluation harness.

Process flow diagram or Use-case diagram

LIVE CALL PATH

1. Call audio
Speakerphone mic, shared Google Meet tab, or demo call. 16 kHz PCM from the browser.

→

2. Gemini analysis
8-second audio slices to Gemini Flash-Lite: transcript + scam signals + exact quotes. (Optional Gemini Live streaming.)

→

3. Risk meter
Calm / Caution / Warning / STOP, escalating with hysteresis. Meet video frame checked for staged-authority cues.

→

4. Act
Instant guardian alert + cool-off hold when money / OTP is asked. Gemini Flash agent: advisory, summary, complaint draft.

Genuine call
No signals: stays Calm. Nothing is raised, nothing is stored.

Warning
Calm whisper in English / Hindi + "Hang up and call family" button.

STOP
Guardian phone alerted (Firestore realtime). Guardian approves the cool-off hold.

Final report
Written from the full call for 1930 / cybercrime.gov.in: amount, timeline, quotes. Never filed automatically.

SCAM CHECK PATH

Input
Screenshot, voice note / call recording, pasted text, or a question. Profile: senior / adult / young adult / teen.

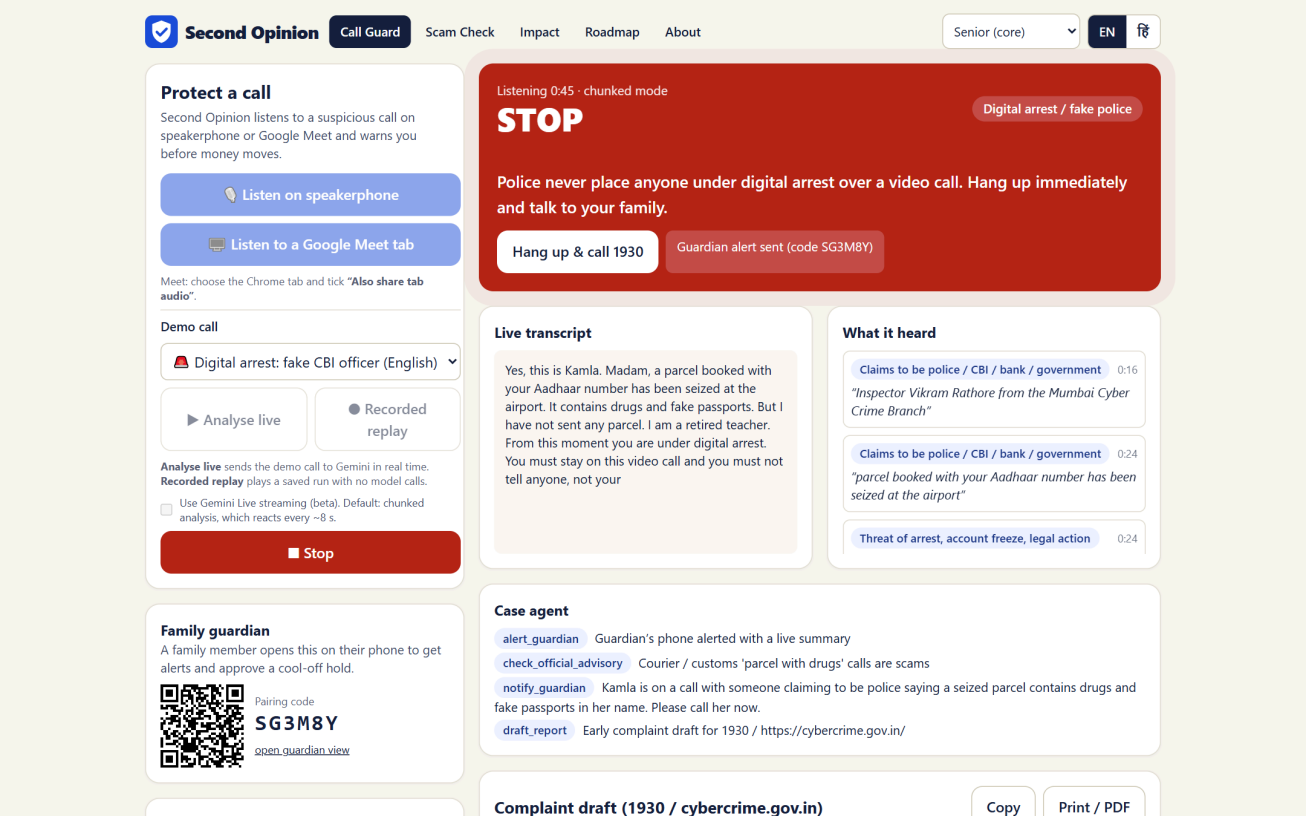
→

Gemini multimodal
Structured JSON verdict: risk 0-100, scam type, red flags with quotes.

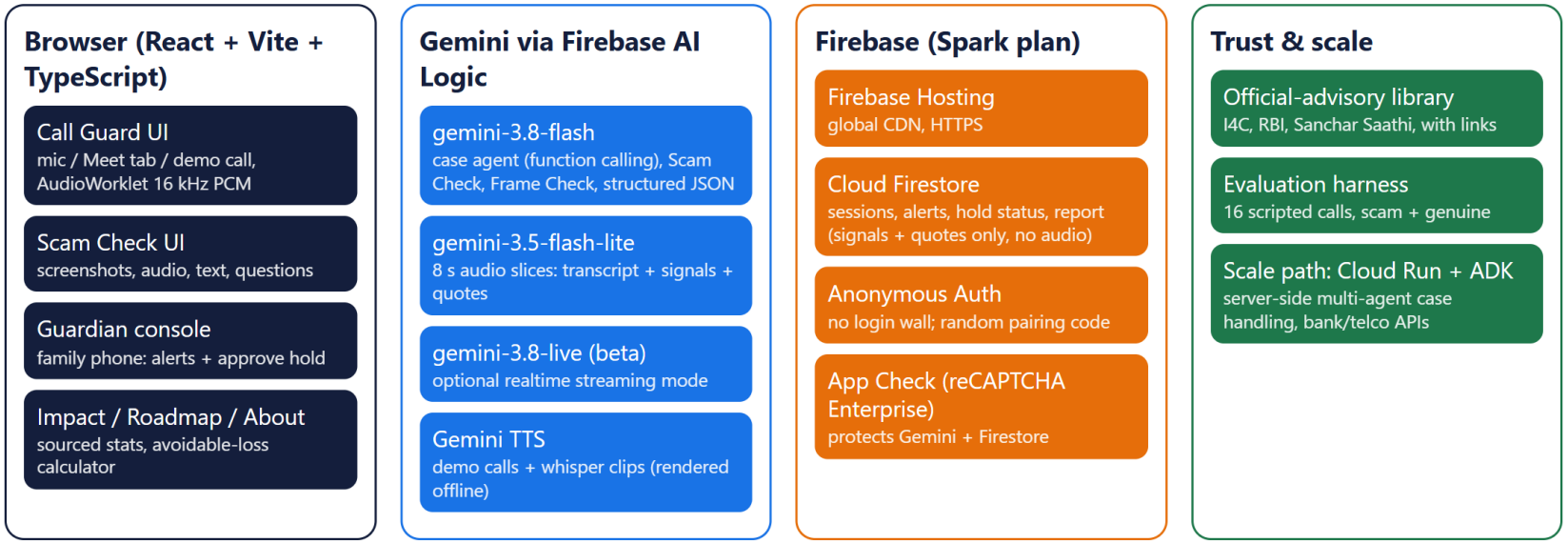
→

Guidance
What to do now, safe reply, official advisory links, helplines, follow-up chat.

Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution



Data flow: browser audio/images → Firebase AI Logic (App Check verified) → Gemini → structured alerts → Firestore → guardian phone. Rules: only the session owner writes; a guardian may change only the hold status.

Technologies to be used in the solution

Google AI and Google Cloud

- **Gemini via Firebase AI Logic:** gemini-3.8-flash (case agent with function calling, final report, Scam Check multimodal + structured JSON, Frame Check vision); gemini-3.5-flash-lite (8-second call-audio slices: transcript + scam signals); gemini-3.8-live (optional realtime streaming, beta); Gemini TTS (demo calls). Automatic fallback to gemini-3.6-flash / 3.5-flash on rate limits or timeouts.
- **Firebase:** Hosting (deployed prototype), Cloud Firestore (realtime guardian sessions), Anonymous Auth, App Check (reCAPTCHA Enterprise).
- **Scale path:** Cloud Run + Agent Development Kit for server-side multi-agent case handling; Agent Platform memory; Gemma 4 on-device.

Application

- React 19, Vite, TypeScript, Tailwind CSS; Web Audio API (AudioWorklet, 16 kHz PCM); getDisplayMedia for Meet tab audio and video frames.
- Python tools (google-genai): demo-call TTS, recorded-replay builder, evaluation harness, submission builder.

Safety and privacy

- Curated official-advisory library (I4C, RBI, Sanchar Saathi) with links; exact quotes for every signal; guardian approval; no raw audio or video stored.
- Firestore rules: only the session owner writes; a guardian can only approve or decline a pending hold.

Estimated implementation cost (optional)

Prototype: USD 0. Firebase Spark plan (Hosting, Firestore, Auth, App Check) and the Gemini API free tier.

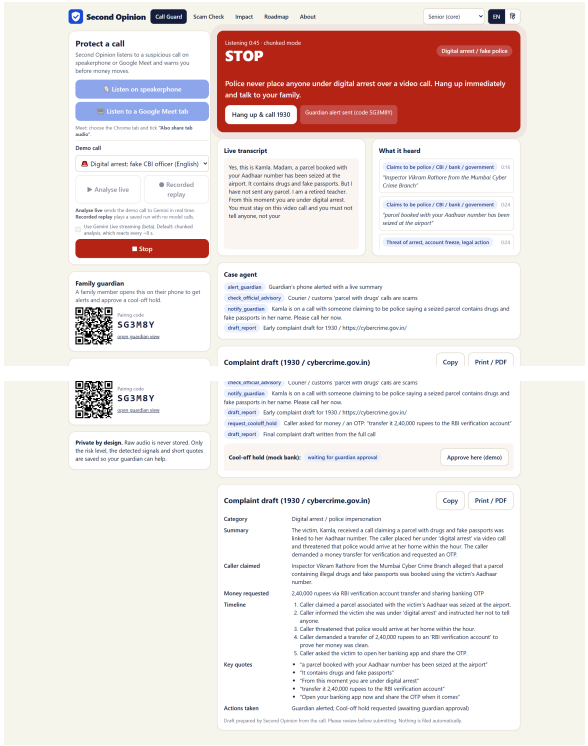
Production estimate (assumptions, not a quote)

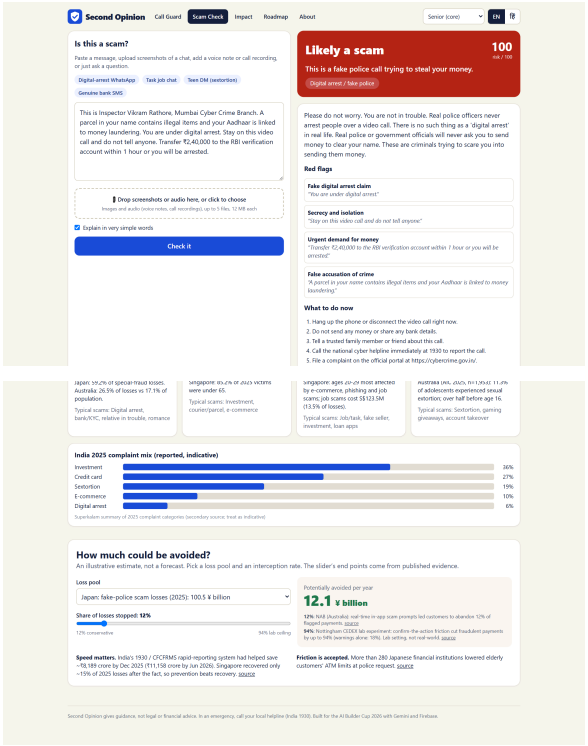
- One protected 10-minute call is ~75 audio slices of 8 s (~860 input and ~150 output tokens each), i.e. ~65k input + ~11k output tokens.
- At Gemini 3.8 Flash intro prices (USD 0.75 / 3.75 per 1M tokens) that is ~USD 0.09 per 10-minute call; Flash-Lite is cheaper.
- Case agent + final report: a few extra calls, ~USD 0.01 per escalated call. Firestore / Hosting: well under USD 0.01 per call.

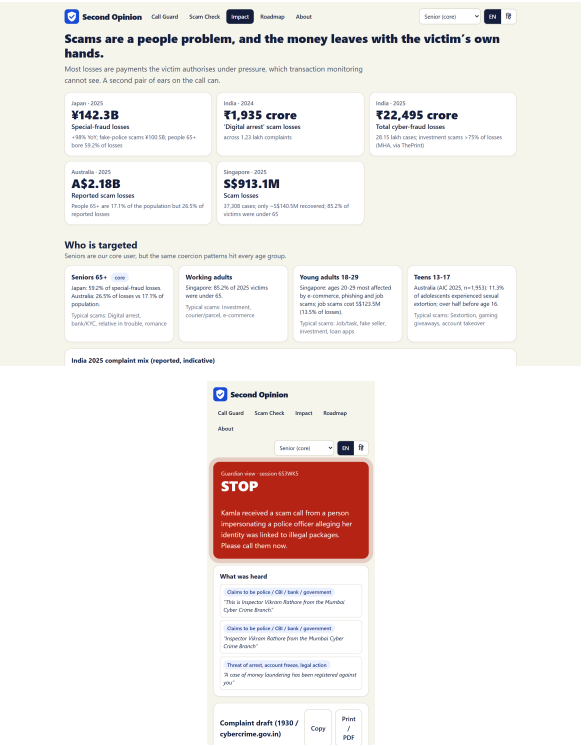
Protecting 10,000 calls a month costs roughly USD 1,000 or less, against an average loss of ~INR 1.6 lakh per digital-arrest complaint (INR 1,935 crore / 1.23 lakh complaints).

Costs fall further with on-device Gemma 4 for first-line detection and server calls only for escalations.

Snapshots of the prototype







Call Guard at STOP · final complaint draft · Scam Check verdict · Impact page · guardian phone view | https://gen-lang-client-0095149296.web.app

Prototype Performance report/Benchmarking

Evaluation (gemini-3.5-flash-lite): 16 scripted calls (English, Hindi, Hinglish, Japanese) through the same classifier prompt the app ships. Scam calls detected: **100% of 10**. False alarms on genuine calls: **0% of 6**. Median time to first warning: **~10 s** (estimated from speaking rate).

Measured end-to-end on the live site (62 s English digital-arrest call): STOP at ~24 s with exact quotes; guardian alerted instantly; cool-off hold requested the moment the transfer was demanded; final complaint draft ~15 s after the call ended. Scam Check: text ~9 s, a 28 s call recording ~40 s.

Call	Type	Peak	1st warning
digital-arrest-en	scam	stop	24s
digital-arrest-hi	scam	stop	20s
digital-arrest-hinglish	scam	stop	10s
bank-kyc-otp	scam	stop	10s
relative-in-trouble	scam	stop	8s
tax-refund	scam	stop	22s
investment-guaranteed	scam	stop	24s
courier-customs	scam	stop	8s

Call	Type	Peak	1st warning
jp-fake-police	scam	stop	6s
jp-oreore	scam	stop	5s
genuine-bank-callback	genuine	calm	-
genuine-doctor	genuine	calm	-
genuine-grandson	genuine	calm	-
genuine-delivery	genuine	calm	-
genuine-insurance-renewal	genuine	calm	-
genuine-electricity	genuine	calm	-

Scripted, small-scale and indicative, not a field trial. Full table: eval/results.md in the repository.

Additional Details/Future Development (if any)

From one grandparent's phone call to every age group

- **Teen Shield:** sextortion, gaming-currency and giveaway scams; non-judgmental tone; risk-only alerts to a trusted adult.
- **Young adults and merchants:** job / task scams, fake sellers, loan apps, fake payment screenshots, UPI collect-request traps.
- **Bank and telco integration:** a real cool-off hold API, call-risk metadata, auto-reporting to Sanchar Saathi / 1930.
- **On-device Gemma 4:** private offline detection, Android call screening, WhatsApp bot, Meet / WhatsApp Web extension.
- **Production backend:** Cloud Run + ADK multi-agent case handling, Agent Platform memory, observability.
- **More JAPAC languages** via Gemini Live translation; Live streaming for sub-second reactions on continuous call audio.

Honest limits today

- Demo calls and speakerphone / Meet audio only (phones block third-party call-audio capture).
- Free-tier Gemini data may be used to improve Google products: production needs paid-tier or Vertex AI data controls and consent flows.
- Evaluation is scripted, not field-tested.

Provide links to your:

- GitHub Public Repository https://github.com/dager23/second-opinion
- Demo Video Link (3 Minutes) (add demo video link)
- Final Product Link https://gen-lang-client-0095149296.web.app

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Thank you

